

Confusion Matrix Example 1

1. Confusion Matrix (Color Highlighted)

True \ Pred	A	B	C
A	2	1	0
B	0	1	1
C	1	1	2

2. Per-Class Metrics

Formulas:

$$\text{Precision}_i = \frac{TP_i}{TP_i + FP_i}, \quad \text{Recall}_i = \frac{TP_i}{TP_i + FN_i}, \quad F1_i = 2 \cdot \frac{\text{Precision}_i \cdot \text{Recall}_i}{\text{Precision}_i + \text{Recall}_i}$$

Class A

$$TP_A = 2, \quad FP_A = 1, \quad FN_A = 1$$

$$\text{Precision}_A = 0.667, \quad \text{Recall}_A = 0.667, \quad F1_A = 0.667$$

Class B

$$TP_B = 1, \quad FP_B = 2, \quad FN_B = 1$$

$$\text{Precision}_B = 0.333, \quad \text{Recall}_B = 0.5, \quad F1_B \approx 0.4$$

Class C

$$TP_C = 2, \quad FP_C = 1, \quad FN_C = 2$$

$$\text{Precision}_C = 0.667, \quad \text{Recall}_C = 0.5, \quad F1_C \approx 0.571$$

3. Macro-Average Metrics

$$\text{Macro Precision} \approx 0.556, \quad \text{Macro Recall} \approx 0.556, \quad \text{Macro F1} \approx 0.546$$

4. Accuracy & Error Rate

$$\text{Accuracy} = 0.556, \quad \text{Error Rate} \approx 0.444$$

Confusion Matrix Example 2

1. Confusion Matrix (Color Highlighted)

True \ Pred	A	B	C
A	3	1	1
B	1	2	1
C	0	1	4

2. Per-Class Metrics

Class A

$$TP_A = 3, \quad FP_A = 1, \quad FN_A = 2$$

$$\text{Precision}_A = \frac{3}{3+1} = 0.75, \quad \text{Recall}_A = \frac{3}{3+2} = 0.6, \quad F1_A = 2 \cdot \frac{0.75 \cdot 0.6}{0.75 + 0.6} \approx 0.667$$

Class B

$$TP_B = 2, \quad FP_B = 2, \quad FN_B = 2$$

$$\text{Precision}_B = \frac{2}{2+2} = 0.5, \quad \text{Recall}_B = \frac{2}{2+2} = 0.5, \quad F1_B = 0.5$$

Class C

$$TP_C = 4, \quad FP_C = 1, \quad FN_C = 1$$

$$\text{Precision}_C = \frac{4}{4+1} = 0.8, \quad \text{Recall}_C = \frac{4}{4+1} = 0.8, \quad F1_C = 0.8$$

3. Macro-Average Metrics

$$\text{Macro Precision} = \frac{0.75 + 0.5 + 0.8}{3} \approx 0.683$$

$$\text{Macro Recall} = \frac{0.6 + 0.5 + 0.8}{3} \approx 0.633$$

$$\text{Macro F1} = \frac{0.667 + 0.5 + 0.8}{3} \approx 0.656$$

4. Accuracy & Error Rate

$$\text{Accuracy} = \frac{TP_A + TP_B + TP_C}{\text{Total}} = \frac{3 + 2 + 4}{3 + 1 + 1 + 1 + 2 + 1 + 0 + 1 + 4} = \frac{9}{14} \approx 0.643$$

$$\text{Error Rate} = 1 - 0.643 \approx 0.357$$

Practical Confusion Matrix Examples with Full Data and Step-by-Step Computations

Example 1: Email Spam Classification

Scenario: Classify emails as Spam (S) or Not Spam (N).

Step 0: Raw Data (True vs Predicted)

Email ID	True Label	Predicted Label
1	S	S
2	N	S
3	N	N
4	S	S
5	N	N
6	S	N
7	S	S
8	N	N
9	N	S
10	S	N

Step 1: Construct Confusion Matrix

- Count TP (True Positives), FP (False Positives), FN (False Negatives), TN (True Negatives):

True \ Pred	S	N
S	3	2
N	2	3

Step 2: Compute Metrics for Spam (S)

$$\text{Precision}_S = \frac{TP}{TP + FP} = \frac{3}{3 + 2} = \frac{3}{5} = 0.6$$

$$\text{Recall}_S = \frac{TP}{TP + FN} = \frac{3}{3 + 2} = \frac{3}{5} = 0.6$$

$$F1_S = 2 \cdot \frac{0.6 \cdot 0.6}{0.6 + 0.6} = 0.6$$

$$\text{Accuracy} = \frac{TP + TN}{Total} = \frac{3 + 3}{10} = 0.6, \quad \text{Error Rate} = 1 - 0.6 = 0.4$$

Example 2: Disease Diagnosis (3-Class)

Scenario: Classify patients as Healthy (H), Diabetes (D), Hypertension (Hy).

Step 0: Raw Data (True vs Predicted)

Patient ID	True Label	Predicted Label
1	H	H
2	D	D
3	Hy	Hy
4	H	D
5	D	D
6	Hy	D
7	H	H
8	D	Hy
9	Hy	Hy

Step 1: Construct Confusion Matrix

True \ Pred	H	D	Hy
H	2	1	0
D	0	2	1
Hy	0	1	2

Step 2: Compute Metrics for Each Class

Class H:

$$TP_H = 2, \quad FP_H = 0 + 0 = 0, \quad FN_H = 1 + 0 = 1$$

$$\text{Precision}_H = \frac{2}{2 + 0} = 1$$

$$\text{Recall}_H = \frac{2}{2 + 1} = \frac{2}{3} \approx 0.667$$

$$F1_H = 2 \cdot \frac{1 \cdot 0.667}{1 + 0.667} \approx 0.8$$

Class D:

$$TP_D = 2, \quad FP_D = 1 + 1 = 2, \quad FN_D = 1$$

$$\text{Precision}_D = \frac{2}{2 + 2} = 0.5$$

$$\text{Recall}_D = \frac{2}{2 + 1} = \frac{2}{3} \approx 0.667$$

$$F1_D = 2 \cdot \frac{0.5 \cdot 0.667}{0.5 + 0.667} \approx 0.571$$

Class Hy:

$$TP_{Hy} = 2, \quad FP_{Hy} = 0 + 1 = 1, \quad FN_{Hy} = 1$$

$$\text{Precision}_{Hy} = \frac{2}{2+1} = \frac{2}{3} \approx 0.667$$

$$\text{Recall}_{Hy} = \frac{2}{2+1} = \frac{2}{3} \approx 0.667$$

$$F1_{Hy} = 2 \cdot \frac{0.667 \cdot 0.667}{0.667 + 0.667} \approx 0.667$$

Step 3: Macro-Average Metrics

$$\text{Macro Precision} = \frac{1 + 0.5 + 0.667}{3} \approx 0.722$$

$$\text{Macro Recall} = \frac{0.667 + 0.667 + 0.667}{3} = 0.667$$

$$\text{Macro F1} = \frac{0.8 + 0.571 + 0.667}{3} \approx 0.679$$

$$\text{Accuracy} = \frac{2 + 2 + 2}{9} = 0.667, \quad \text{Error Rate} = 0.333$$

Example 3: Sentiment Analysis (3-Class)

Scenario: Predict sentiment of reviews: Positive (P), Neutral (N), Negative (Neg).

Step 0: Raw Data (True vs Predicted)

Review ID	True Label	Predicted Label
1	P	P
2	P	N
3	N	N
4	Neg	Neg
5	P	P
6	N	P
7	Neg	Neg
8	P	Neg
9	N	N
10	Neg	N
11	P	P
12	N	N

Step 1: Construct Confusion Matrix

True \ Pred	P	N	Neg
P	3	1	1
N	1	3	0
Neg	0	1	2

Step 2: Compute Metrics for Each Class

Class P:

$$TP_P = 3, \quad FP_P = 1 + 0 = 1, \quad FN_P = 1 + 1 = 2$$

$$\text{Precision}_P = \frac{3}{3 + 1} = 0.75$$

$$\text{Recall}_P = \frac{3}{3 + 2} = 0.6$$

$$F1_P = 2 \cdot \frac{0.75 \cdot 0.6}{0.75 + 0.6} \approx 0.667$$

Class N:

$$TP_N = 3, \quad FP_N = 1 + 1 = 2, \quad FN_N = 1$$

$$\text{Precision}_N = \frac{3}{3 + 2} = 0.6$$

$$\text{Recall}_N = \frac{3}{3 + 1} = 0.75$$

$$F1_N = 2 \cdot \frac{0.6 \cdot 0.75}{0.6 + 0.75} \approx 0.667$$

Class Neg:

$$TP_{Neg} = 2, \quad FP_{Neg} = 1 + 0 = 1, \quad FN_{Neg} = 1$$

$$\text{Precision}_{Neg} = \frac{2}{2 + 1} \approx 0.667$$

$$\text{Recall}_{Neg} = \frac{2}{2 + 1} \approx 0.667$$

$$F1_{Neg} = 2 \cdot \frac{0.667 \cdot 0.667}{0.667 + 0.667} \approx 0.667$$

Step 3: Macro-Average Metrics

$$\text{Macro Precision} = \frac{0.75 + 0.6 + 0.667}{3} \approx 0.672$$

$$\text{Macro Recall} = \frac{0.6 + 0.75 + 0.667}{3} \approx 0.672$$

$$\text{Macro F1} = \frac{0.667 + 0.667 + 0.667}{3} = 0.667$$

$$\text{Accuracy} = \frac{3 + 3 + 2}{12} = 0.667, \quad \text{Error Rate} = 0.333$$